

Amrut Prajapati

AI/ML Engineer · Ahmedabad, India

Email amratprajapati18123@gmail.com

Portfolio amrut-prajapati.github.io

GitHub github.com/Amrut-Prajapati

LinkedIn linkedin.com/in/amrut-prajapati

Based in Ahmedabad, India · remote-ready

5,000+

automated tests

on a real-time video analytics platform

< 20 ms

per frame

real-time 3D mapping

₹2,00,000

scholarship

_VOIS Innovation Marathon 2.0 winner

7B

vision-language model

fine-tuned and run on a single GPU

PROFILE

AI/ML engineer who builds computer-vision, vision-language and agentic systems that keep working after the demo: held-out testing, thousands of automated tests, multi-hour soak runs, and models engineered for modest hardware. Winner of the ₹2,00,000 _VOIS Innovation Marathon 2.0 with ReUpyog, a confidential flagship project. Final-year B.Tech CSE, GLS University.

Open to AI/ML engineering roles (full-time, contract or internship), early-stage teams and select freelance builds, remote or Ahmedabad.

WHAT I BUILD

Computer vision

Detection, segmentation, tracking and OCR, evaluated on held-out data.

3D perception and robotics

Point-cloud perception, real-time mapping and robotics-middleware integration.

Operator consoles

React and Next.js dashboards with live video, 3D and map views.

Vision-language and LLM systems

Fine-tuned models, tool-using assistants and retrieval, with evidence and confidence on every answer.

Production AI backends

FastAPI services, PostgreSQL and Redis, Docker, ONNX and TensorRT optimisation.

Quality engineering

Automated tests, property-based testing and multi-hour soak runs.

THE STANDARD I BUILD TO

- **Quality.** Held-out test sets, not cherry-picked demos. Results reported as measured, with the limits stated.
- **Scalability.** Pipelines sized for the hardware you have: real-time systems that run on a single GPU.
- **Reliability.** Thousands of automated tests and multi-hour soak runs before anything is called done.
- **Impact.** Built for India's real problems: smarter cities, sustainability, national growth.

FLAGSHIP PROJECT

ReUpyog: A confidential AI project

SSIP seed grant, Government of Gujarat · Capstone project (I & II, both O+), GLS University

- Winner of the ₹2,00,000 _VOIS Innovation Marathon 2.0.
- SSIP seed grant, Government of Gujarat.
- Capstone Project I and II at GLS University, both graded O+.
- Technical details are confidential and available to serious enquiries on request.

SELECTED BUILDS · DETAILS CONFIDENTIAL

Real-time video analytics platform

A GPU-accelerated platform with a full-stack operator console, engineered to production standards: proven under multi-hour soak tests and 5,000+ automated tests.

Real-time 3D perception & mapping system

Segmentation network trained from scratch; GPU-accelerated mapping under 20 ms per frame; 3D web console.

Agentic vision-language assistant

A fine-tuned 7B vision-language assistant on a single GPU that routes questions to specialist vision tools and backs every answer with evidence and a confidence score.

Technology-intelligence & forecasting platform

Smart India Hackathon 2025 finalist. Automates the collection, analysis and long-range forecasting of strategic-technology trends, with an offline GPU-powered AI engine, a secured API and an executive dashboard with one-click report export.

RECOGNITION

Winner, _VOIS Innovation Marathon 2.0 Feb 2026
₹2,00,000 scholarship for ReUpyog. Presented at the _VOIS 2.0 National Innovation Showcase. Programme by Vodafone Idea Foundation and Edunet Foundation.

SSIP seed grant, Government of Gujarat Mar 2026
Certificate of Appreciation from the GLS University Incubation Centre for outstanding innovation, with a seed grant awarded for the ReUpyog project.

Finalist, Smart India Hackathon 2025 2025
DRDO problem domain. Certificate of Appreciation from GLS University.

EDUCATION

B.Tech, Computer Science & Engineering 2023 – 2027 (expected)
GLS University, Faculty of Engineering & Technology, Ahmedabad
CGPA 7.95 / 10 · SGPA 8.4+ in each of the last three semesters

COURSEWORK WITH GRADES (O+ OUTSTANDING, O EXCELLENT)

Artificial Intelligence	O+
Machine Learning with Lab	O+ / O+
Digital Image & Video Processing with Lab	O / O+
Recommender Systems	O
Probability & Statistics	O
Discrete Mathematics	O+
Linear Algebra	A
Data Structures	O
Design & Analysis of Algorithms	A
RDBMS	O
Cloud Computing & Virtualization	O
Cryptography & Information Security	O
Capstone Project I & II	O+ / O+

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional · Oracle, Oct 2025
Oracle Cloud Infrastructure 2025 Certified Data Science Professional · Oracle, Oct 2025
Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate · Oracle, Oct 2025
Core Java Programming · GLS University MOOC, grade O+

TECHNICAL SKILLS

Computer vision & 3D: YOLO family, RT-DETR, SAM2, OpenCV, semantic segmentation, multi-object tracking, OCR, LiDAR and point clouds

Vision-language & LLMs: Fine-tuning (QLoRA / PEFT), tool-calling agents, RAG, local LLMs, prompt engineering, Whisper, diffusion models

Machine learning: PyTorch, TensorFlow, deep learning, model evaluation

Deployment: ONNX, TensorRT, CUDA, FP16 optimisation, Docker, ROS 2

Backend & data: FastAPI, REST, PostgreSQL / PostGIS, Redis, Streamlit, Gradio, n8n

Frontend: React, Next.js, TypeScript, Three.js, MapLibre

Testing: pytest, Hypothesis, Vitest, Playwright

Languages: Python, Java, C, C++, JavaScript, TypeScript, SQL · English, Hindi, Gujarati

HOW I WORK

- **1. Scope.** A short call and a written scope with acceptance tests, so “done” is defined before work starts.
- **2. Build.** Weekly demos on real data, not slides.
- **3. Verify.** Held-out evaluation, automated tests and soak runs. You get the numbers, including the limits.
- **4. Hand over.** Documented, tested code and deployment notes. You own it.
- **5. Support.** Clear handover notes, and questions answered after delivery.

Replies within one working day · Works in India Standard Time · Trains and tests on a local 32 GB RTX 5090 workstation · AI coding agents in the workflow; every line reviewed and tested.

REFERENCES

Available on request.